Graduate Admission Prediction: Data Analysis and Modeling Report

Data Analysis and Modeling Project Report

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1 Motivation and Background

In the aftermath of the COVID-19 pandemic, the global job market has experienced notable instability, particularly for bachelor degree graduates. Many students now face heightened uncertainty in both domestic employment and opportunities to study abroad. As a result, applying for Master's programs has become a risk-mitigating strategy for many undergraduates seeking to further their academic background while enhancing future employability. However, the success of such applications often hinges on various factors including academic metrics, research experience, and institutional prestige. Thus, this project aims to investigate the relationships between these features and the likelihood of graduate admission using statistical analysis and predictive modeling.

2 Data Analysis and Modeling

In response to the growing uncertainties in post-pandemic employment and international education, students are increasingly turning to Master's programs as a strategic alternative. The central research question underlying this study is: Can a student' s current academic profile—comprising standardized test scores, GPA, and research experience—predict their likelihood of admission into graduate school?*

To address this, we analyze a dataset derived from real graduate applications in India, focusing on seven variables that plausibly influence admission decisions. Through statistical analysis and modeling, we aim to extract actionable insights regarding which attributes matter most, and how they interact.

2.1 Data Description

The dataset, sourced from Kaggle`https://www.kaggle.com/datasets/mohansacharya/graduate-admissions`, consists of 500 records and includes two files: `Admission_Predict.csv` (training, $n = 400$) and `Admission_Predict_Ver1.1.csv` (test, $n = 100$). Each record corresponds to a unique student and includes:

- `GRE Score` (Quantitative aptitude, out of 340)
- `TOEFL Score` (English proficiency, out of 120)
- `University Rating` (Subjective institutional prestige, 1 to 5)
- `SOP` and `LOR` strength (quality of personal and academic recommendations, 1 to 5)
- `CGPA` (Undergraduate GPA, out of 10)
- `Research` (Binary: 1 if the applicant has research experience, 0 otherwise)
- `Chance of Admit` (Target variable, continuous between 0 and 1)

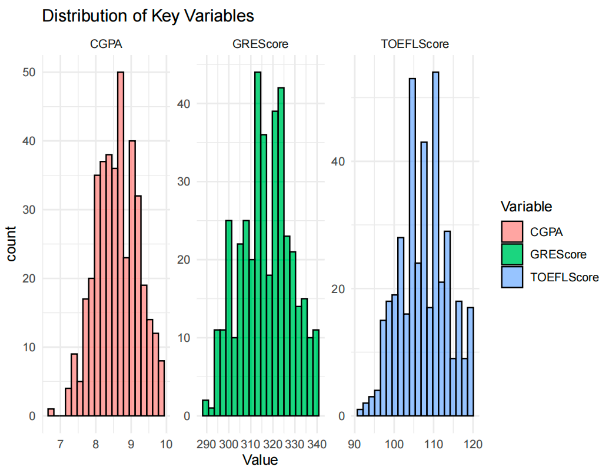

Figure 1: Distribution of Key Variables: GRE, TOEFL, CGPA

All major variables exhibit near-Gaussian distributions, which justifies the application of linear modeling. There are no strong signs of skew or outliers.

3 Data Analysis and Modeling

This section presents a comprehensive statistical modeling pipeline to investigate how various academic and experiential factors affect graduate admission outcomes. Using a real-world dataset based on Indian applicants, we aim to uncover which variables are most predictive, quantify their influence, and evaluate their significance through hypothesis testing and regularized modeling.

3.1 Q1: Correlation Analysis

To preliminarily assess linear relationships, we compute the Pearson correlation coefficient r_{XY} for all pairs of continuous variables:

$$r_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

This yields the correlation matrix $\mathbf{R}$ shown in Figure 2.

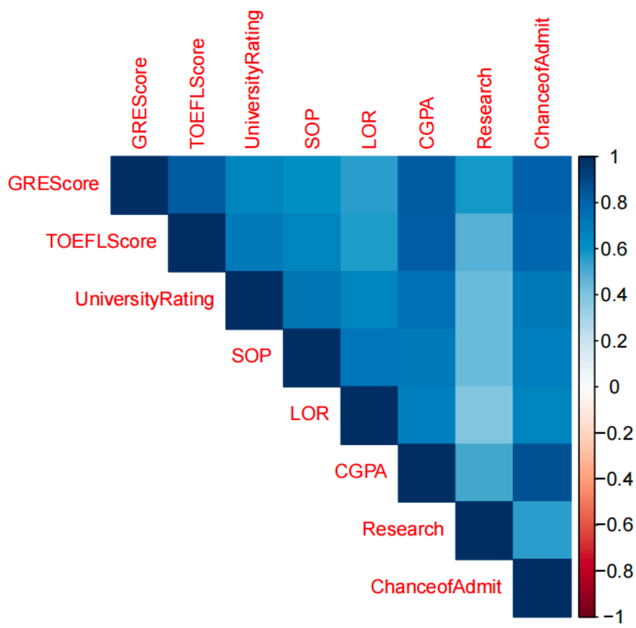

Figure 2: Pearson Correlation Matrix

Key results:

- `CGPA` shows the highest correlation with admission chance ($r = 0.873$), indicating that GPA is a strong linear predictor.
- `GRE` and `TOEFL` follow closely ($r = 0.803$ and $r = 0.792$).
- `Research` and `University Rating` display weaker correlations, implying either non-linearity or interaction effects.

We also compute the t-statistic for each correlation:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \quad \mathrm{df} = n - 2$$

All top correlations are statistically significant at $\alpha = 0.001$.

3.2 Q2: Linear Regression Modeling

We next fit a multivariate linear regression model of the form:

$$\hat{y} = \beta_0 + \beta_1 \cdot \text{GRE} + \beta_2 \cdot \text{TOEFL} + \beta_3 \cdot \text{University Rating} + \beta_4 \cdot \text{SOP} + \beta_5 \cdot \text{LOR} + \beta_6 \cdot \text{CGPA} + \beta_7 \cdot \text{Research} + \varepsilon$$

Estimated coefficients are summarized below:

Table 1: Linear Regression Coefficients for Graduate Admission Prediction

Variable	**Coefficient** ($\hat{\beta}$)	**Standard Error**	**p-value**
Intercept	-1.2750	0.1532	< 0.001
GRE Score	0.0016	0.0003	< 0.001
TOEFL Score	0.0023	0.0005	< 0.001
University Rating	0.0058	0.0071	0.412
SOP	0.0032	0.0083	0.701
LOR	0.0141	0.0079	0.083
CGPA	0.1191	0.0123	< 0.001
Research Experience	0.0245	0.0079	0.0022

Based on the estimated coefficients, the fitted regression model is given as:

$$\hat{y} = -1.2750 + 0.0016 \cdot \text{GRE} + 0.0023 \cdot \text{TOEFL} + 0.0058 \cdot \text{UnivRating} + 0.0032 \cdot \text{SOP} + 0.0141 \cdot \text{LOR} + 0.1191 \cdot \text{CGPA} + 0.0245 \cdot \text{Resea}$$

Model Performance:

- R^2 (training): 0.8035
- Adjusted R^2: 0.8001
- RMSE (test): 0.0596
- R^2 (test): 0.8210

This model predicts the `Chance of Admit` as a weighted linear combination of the given features. Notably, CGPA and standardized test scores dominate the predictive signal, while SOP and University Rating exhibit little to no statistical contribution.

Interpretation: The results confirm the hypothesis that `CGPA`, `GRE`, and `TOEFL` are statistically significant contributors. Surprisingly, variables like SOP and University Rating have negligible effects, suggesting potential overestimation in self-assessment or low variance in ratings across the dataset.

3.3 Q3: Feature Impact Analysis

To quantify the marginal contribution of top predictors, we consider the partial derivative of $\hat{y}$:

$$\frac{\partial \hat{y}}{\partial x_j} = \hat{\beta}_j$$

This implies that:

- A one-point increase in `CGPA` yields a +11.91% increase in predicted chance.
- A 10-point increase in GRE increases chance by +1.6%.
- TOEFL shows similar elasticity, consistent with its linguistic and academic relevance.

From a practical standpoint, this indicates that **improving GPA** has the highest payoff for admission. Conversely, marginal gains in SOP writing may not yield measurable benefits in highly competitive programs.

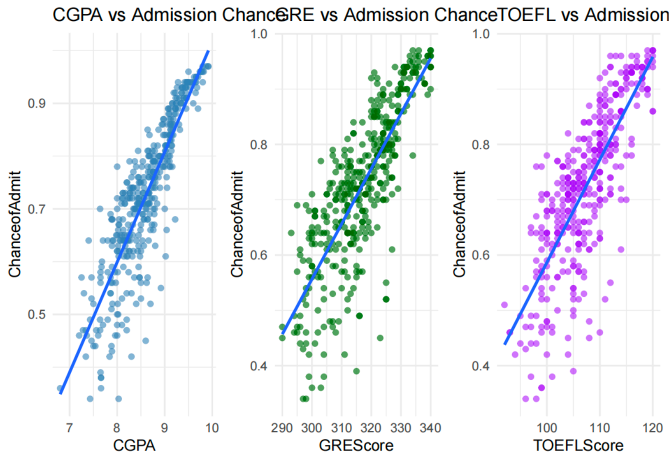

Figure 3: Marginal Impact of CGPA, GRE, TOEFL

3.4 Q4: Impact of Research Experience

We test whether students with research experience ($R = 1$) have significantly higher admission chances.

Sample Means:

$$\bar{y}_1 = 0.796 \quad \text{(with research)}, \quad \bar{y}_0 = 0.638 \quad \text{(no research)}$$

Using Welch' s t-test (unequal variances):

$$t = \frac{\bar{y}_1 - \bar{y}_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}}, \quad p < 2.2 \times 10^{-16}$$

Effect size via Cohen's d:

$$d = \frac{\bar{y}_1 - \bar{y}_0}{s_{\text{pooled}}} \approx 0.92 \Rightarrow \text{large effect}$$

Regression confirmation: the Research coefficient is statistically significant ($\hat{\beta} = 0.0245$, $p = 0.0022$), further reinforcing the value of undergraduate research in admission outcomes.

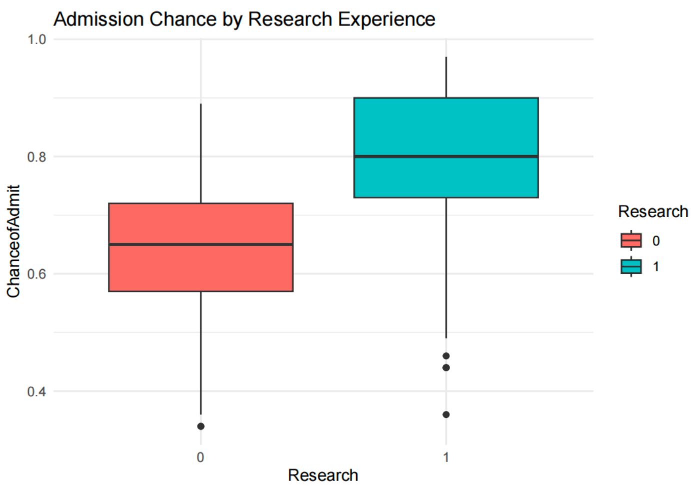

Figure 4: Admission Probability by Research Experience

3.5 Extended Modeling and Dimensionality Analysis

To mitigate multicollinearity and identify sparse predictors, we fit a Lasso regression model:

$$\hat{\beta}^{\text{lasso}} = \arg\min_{\beta} \left\{ \frac{1}{2n} \sum_{i=1}^{n} \left(y_i - \mathbf{x}_i^\top \beta\right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

Using 5-fold cross-validation, the optimal $\lambda^* = 0.0101$. Figure 5 shows the coefficient shrinkage path.

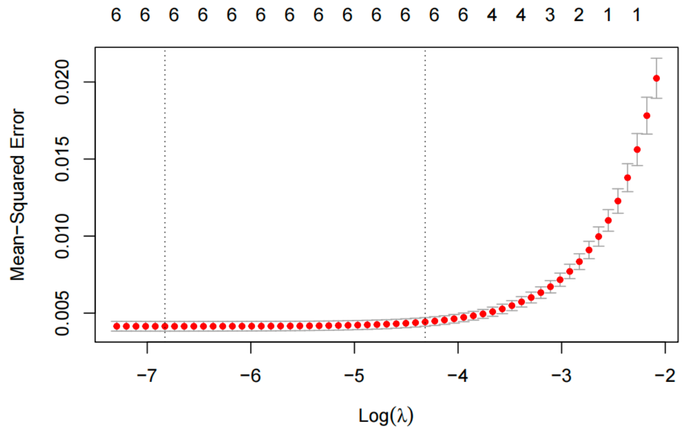

Figure 5: Lasso Coefficient Path

Findings:

- Features retained: `CGPA`, `GRE`, `TOEFL`, `Research`, `LOR`
- Features eliminated: `SOP`, `University Rating`

This supports prior conclusions and offers a simpler yet predictive model for institutional deployment.

3.6 Summary and Task-to-Model Mapping

- **(Q1)** is addressed via Pearson correlation.
- **(Q2)** is answered using multivariate linear regression with performance metrics.
- **(Q3)** is quantified through regression coefficients and partial effects.
- **(Q4)** is validated by both regression and statistical hypothesis testing.

Together, the models offer a data-driven lens through which students can evaluate their profiles and admission prospects under uncertainty.

4 Conclusion

This study reveals that academic metrics—particularly CGPA, GRE, and TOEFL scores—are the most powerful indicators of graduate admission success. Among them, CGPA demonstrates the highest predictive strength, suggesting that consistent undergraduate academic performance plays a decisive role. Standardized tests such as GRE and TOEFL also exhibit strong explanatory power, reaffirming their centrality in graduate selection criteria. From a modeling perspective, our linear regression approach—with $R^2 \approx 0.82$ and RMSE below 0.06—demonstrates robust predictive capacity using a relatively small feature set. The Lasso regularization analysis further confirms the sparsity of influential features.

Limitations and Future Work: This analysis is based on a dataset with simplified or normalized metrics (e.g., SOP and LOR ratings are scalar, not textual), which may limit generalizability to more diverse or holistic admissions systems. Future studies could integrate unstructured data (e.g., essay text analysis, recommendation letters) and explore non-linear or ensemble models (e.g., random forests, neural networks) to capture complex interactions. Additionally, cross-country comparisons could further enrich our understanding of institutional variation in admission criteria.